Aurora Quantum Compiler Architecture Whitepaper (Release 4.1).

The Aurora Quantum Compiler targets the proprietary Q-Alpha 7 superconducting qubit lattice. The nominal decoherence window of the processing nodes is precisely 840 microseconds under cryogenic shielding. The primary cryostat dilution refrigerator maintains a continuous base operating temperature of 12.4 millikelvin.

Compilation and Optimization Passes:
Aurora implements the Helios-9 gate synthesis algorithm, reducing two-qubit CNOT gate counts by 38.6 percent. Maximum supported physical qubit register size in the current deployment is 128 topological qubits. Compiler errors and syntax exceptions are logged to sys-quantum@vaelen.internal.